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ORIGINAL ARTICLE

Assessment of superior vena cava flow and cardiac output in different patterns of patent ductus arteriosus shunt

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Abstract

Background: Despite the widespread use of superior vena cava (SVC) flow as a marker of systemic blood flow from the upper body, no previous studies have systematically evaluated the correlation between SVC flow and other echocardiography measures of systemic blood flow in the context of different patterns of patent ductus arteriosus (PDA) shunt direction

Methods: A retrospective cohort study of preterm infants (< 30 weeks, < 21 days of life) who underwent comprehensive targeted neonatal echocardiography (TnECHO) was performed. Patients were categorized as follows: (i) Hemodynamically significant left-to-right shunt; (ii): Bidirectional shunt; (iii) No PDA or insignificant shunt. SVC flow, as measured by two distinct methods, was compared to left and right ventricular outputs (LVO and RVO). Intra- and inter-observer reliability testing was performed

Results: In total, 45 patients were included (15 in each group) with a median [IQR] weight of 720 [539, 917] grams at the time of assessment. SVC dimensions and flow measurements were not different between the groups, although patients with left-to-right shunt had higher LVO/RVO ratio. SVC flow, as estimated using the modified method, had a strong correlation with LVO ($r = .63$, $p = 0.012$) and RVO ($r = .635$, $p = 0.011$) in patients with no PDA. Inter- and intra-observer reliability were both stronger for LVO and RVO when compared to SVC flow measurements

Conclusion: SVC flow was comparable across all three groups irrespective of higher LVO and LVO/RVO ratio in patients with left-to-right shunts. This may reflect poor measurement reliability or compensation for left-to-right ductal shunt by increased LVO to maintain systemic perfusion.

KEYWORDS
left ventricular output, patent ductus arteriosus, superior vena cava flow, targeted neonatal echocardiography

1 | INTRODUCTION

Neonates, particularly preterm infants, are at risk for systemic and cerebral hypoperfusion due to maladaptive physiologic changes during the transitional period. However, clinical methods of assessing adequacy of the circulation, such as blood pressure, correlate poorly with systemic perfusion.[1] Echocardiography derived estimates of left and right ventricular outputs (LVO and RVO) have been used to assess the adequacy of systemic blood flow but, due to the common presence of shunts (e.g., patent foramen ovale [PFO] and patent ductus arteriosus [PDA]), may not reflect true systemic perfusion.[2] In an attempt to mitigate such confounders, Kluckow et al. proposed that superior vena cava (SVC) flow is a reliable marker of systemic blood flow from the upper body, including the brain,[3] that is not influenced by the presence of shunts. One survey indicated that nearly 40% of clinicians performing functional echocardiography rely on SVC flow measurements to identify circulatory failure.[4] Additionally, several studies reported an association between low SVC flow and intraventricular hemorrhage, adverse neurodevelopmental outcomes, cerebral tissue oxygenation and death in immature infants.[2,5–7]

Subsequent studies have correlated the ratio between LVO and SVC flow as a marker of ductal significance, given that larger shunts have been associated with higher LVO to SVC flow ratio.[5,8] Left-to-right ductal shunt volume increased LVO without significantly impacting upper body flow (SVC) volumes by cardiac magnetic resonance imaging (cMR) and echocardiography.[9,10] Additionally, the traditional method for SVC flow measurement has been challenged[11,12] due to poor correlation with more precise methods such cMR.[13] These mainly relate to intra- and inter-observer reliability because of difficulties with proper alignment of the flow and measurement of SVC diameter.[12] In 2017, a modified echocardiography approach offered an improvement in accuracy and repeatability.[14] Despite the widespread use of SVC flow as a marker of systemic blood flow from the upper body, no previous studies have systematically evaluated the impact of directionality of PDA shunt on the correlation between other measures of systemic blood flow (i.e., LVO and RVO) and SVC flow. It is important to recognize that SVC flow is not a direct substitute for total body blood flow. As the brain is preferentially perfused by pre-ductal cardiac output, we hypothesized that in the presence of a left-to-right PDA shunt, the expected rise in LVO would lead to a consequential increase in SVC flow. The objective of this study was to investigate the relationship between LVO, RVO, and SVC flow, measured concurrently, in preterm infants according to directionality of PDA shunt.

2 | METHODS

We conducted a retrospective cohort study of preterm infants admitted to the neonatal intensive care unit at the University of Iowa Stead Family Children's Hospital, between September 2018 and October 2019. Infants were selected from a database derived from a prospective targeted neonatal echocardiography (TnECHO) Quality Improvement initiative which included imaging of two methods of assessing SVC flow and dimensions. Infants were included if they were less than 30 weeks gestational age at birth and the TnECHO had been performed in the first 21 days of life. The study was approved by the institutional research ethics board with waiver of consent.

2.1 | Data collection

Clinical data were collected from electronic medical charts. The following information were collected: (i) neonatal demographics, including gender, gestational age, birthweight, postmenstrual age, and weight at the time of TnECHO, (ii) ventilation mode and parameters; and (iii) blood gas results within 6 hours of TnECHO assessment.

2.2 | Echocardiography assessment

Comprehensive TnECHO was performed by neonatologists with hemodynamic expertise as part of routine clinical care at the request of the attending neonatologist. All evaluations were performed by neonatologists who completed comprehensive training in image acquisition and interpretation.[15] All TnECHO evaluations were performed according to a standardized protocol which includes approximately 80–100 images and takes up to 15–20 minutes. If this was a first echocardiogram, the assessment included the necessary imaging to screen for congenital heart disease and underwent timely review by pediatric cardiology.[16] All echocardiography analyses were performed, using a dedicated workstation (EchoPAC version BT10; GE Medical Systems), by a single trained investigator (AB). Left/right ventricular outputs and SVC flow were evaluated concurrently (less than 15 min between acquisition of images). A standardized protocol sequence was used for all patients.

2.3 | Superior vena cava flow, traditional method

Superior vena cava diameter was assessed from a high parasternal long axis view, rotated towards the true sagittal plane. The transducer head was placed as close to the midline as possible to acquire directly anteroposterior views of the SVC. Maximum and minimum SVC diameters were assessed in 2D for two consecutive cardiac cycles, and the mean was calculated.[17] The cross-sectional area (CSA) was calculated as [SVC CSA = SVC radius^2 $\times$ π]. SVC flow velocity time integral (VTI) was obtained using pulsed wave (PW) Doppler from a low subcostal view.[3] Any reversal of flow in the SVC was quantified and deducted from the measured forward flow to give an integrated VTI.[17] Five consecutive beats (minimum four) were used to average VTI and heart rate for the PW Dopplers of the SVC. Traditional SVC flow was calculated using a parasternal long axis mean SVC diameter to calculate SVC CSA and the VTI/heart rate from a subcostal view [SVC flow (ml/min/kg) = VTI (cm) $\times$ heart rate (bpm) $\times$ SVC CSA (cm^2)/ weight (kg)].

2.4 | Superior vena cava flow, modified method

A second method of SVC CSA was obtained from a suprasternal view by tracing maximum and minimum internal area of the vessel in two consecutive cardiac cycles, and the average of all measurements was calculated. SVC VTI was also obtained using PW Doppler from supra-sternal view.[14] Modified SVC flow was calculated using suprasternal SVC CSA, VTI, and heart rate from suprasternal view.[14]

2.5 | Left and right ventricular outputs

LVO was obtained by placing a PW Doppler perpendicular to the aortic valve in the apical five-chamber view, with the angle of insonation parallel to the left ventricular outflow tract. No angle correction software was performed and only images with less than 5 degrees deviation from the direction of flow were used. Three consecutive cardiac cycles were measured and averaged. The area under the wave form of the aortic systolic beat was traced to obtain the VTI and the heart rate. The annulus of the aortic valve was measured, from the parasternal long axis, between hinge points with the valve open at the end of ejection. LVO (expressed in ml/min/kg) was calculated by multiplying the aortic CSA [calculated as: (aortic radius^2 $\times$ π)] multiplied by VTI and heart rate and indexed to weight (kg).[13,18] RVO was measured using a similar approach. The right ventricular VTI was obtained by placing PW Doppler at the level of the pulmonary artery valve either in the parasternal short or long axis, whichever view was more parallel to the direction of blood flow (angle of insonation with less than 5 degrees deviation from the direction of flow). Pulmonary artery diameter was measured in the same view, between hinge points, at the end of flow ejection. In the presence of diastolic turbulence due to left-to-right PDA shunt, attempt was made to trace the main envelope of the RVO, excluding the diastolic flow disturbance. We anticipated that intra- and inter-observer reliability would be worse for cross-sectional area/ diameter of the various vessels. Therefore, we performed modified flow measurements by multiplying VTI and heart rate and indexing to weight (without including cross sectional area) for the LVO, RVO and both methods of SVC flow.

Other TnECHO measurements: Additional measurements (i.e., pulmonary venous flow, mitral inflow, measurements of heart function) were performed according to published standards.[18–20] Left ventricular (LV) dysfunction was defined as an ejection fraction (EF) < 55% determined by Simpsons biplane.[21] Moderate to severe right ventricular (RV) dysfunction was defined as tricuspid annular plane systolic excursion (TAPSE) less than two standard deviations of normative data for gestational age.[22] Flow across other shunts (i.e., ventricular septal defect [VSD] and atrial communication) was identified using color Doppler imaging. Low LVO was defined as < 150 ml/min/kg, high LVO was defined as > 250 ml/min/kg and low SVC flow was defined as < 60 ml/min/kg.[5,23]

2.6 | Patent ductus arteriosus evaluation and group classification

The PDA internal diameter was measured at the narrowest portion using two-dimensional imaging. The pattern of flow was assessed with both color and pulsed wave Doppler. Bidirectional ductal shunts were assessed according to proportion of time of left-to-right and right-to-left flow, based on Doppler tracing. PDA score was adapted from Rios et al.,[23] including PDA size indexed to weight, mitral valve E-wave velocity, pulmonary vein D-wave velocity, isovolumic relaxation time, LVO, ratio of left atrium to aorta size, and diastolic flow pattern in the descending aorta, celiac artery and/or middle cerebral artery (MCA) [Table S1]. The maximum score was 14 points. PDA was considered hemodynamically significant if the diameter indexed to weight was at least 1.5 mm/kg and PDA score was at least eight. PDA was considered insignificant if diameter was less than 1 mm/kg, less than .5 mm absolute size and PDA score was less than five. Patients were divided into three groups:

Group 1: Left-to-right PDA shunts which were considered hemodynamically significant and in which medical therapy was recommended.

Group 2: Bidirectional PDA shunt with at least 10% of time spent right-to-left.[24]

Group 3: No transductal shunt or insignificant PDA for which treatment was not indicated.

2.7 | Outcomes

The primary outcome was correlation between SVC flow (traditional or modified) and LVO in patients with significant left-to-right PDA shunt. Secondary outcomes included: (i) correlation between various methods to estimate systemic blood flow for the entire cohort, each subgroup of PDA directionality and patients with no PDA; (ii) correlation between low SVC flow and low LVO; (iii) intra- and inter-rater reliability of measurements; and (iv) correlation between SVC flow and other markers of PDA significance (i.e., PDA score and size).

2.8 | Intra- and inter-observer reliability testing

Reliability testing was performed for a single set of defined images (analysis-reanalysis). Although test-retest evaluation is a superior methodology, we were concerned that the dynamic nature of blood flow and vascular resistance in extremely preterm infants with a PDA would further complicate the analyses. Intra-observer reliability testing was performed for five patients randomly selected from each group (n = 15 total) and inter-observer reliability testing was performed for all patients by a second member of the investigative team.

2.9 | Statistical analysis

Descriptive statistics were used for demographics and clinical data. The Shapiro-Wilk test was used to test continuous variables for normality. Mean (standard deviation) and median (interquartile range) were calculated for data with normal and non-normal distribution, respectively. Continuous variables were compared between the three groups according to PDA shunt pattern using parametric (one-way ANOVA) and non-parametric tests (Kruskal-Wallis) as appropriate. PDA size and PDA modified score were compared with parametric (Student t-test) and non-parametric tests (Mann-Whitney) as appropriate between groups 1 and 2. Categorical variables were presented as frequencies (%) and compared using the $\chi 2$ or Fisher's exact test. Pearson's correlation coefficient was calculated to compare measures of systemic blood flow in the different groups. Intra- and inter-observer reliability testing

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TABLE 1 Demographics, baseline clinical, and echocardiographic characteristics in patients with different patterns of ductal shunt

	Cohort ($n = 45$)	PDA left-to-right ($n = 15$)	PDA bidirectional ($n = 15$)	No PDA ($n = 15$)	p
Male	25 (55.6)	6 (40)	9 (60)	15 (100)	0.061
Birth weight (grams)	720 [539, 890]	716 [515, 910]	725 [565, 865]	826 [513, 980]	NS
Gestational age (weeks)	25.45±2.11	25.77±2.18	25.06±2.17	25.53±2.07	NS
Age at the time of assessment (days)	1 [0, 4]	3 [1, 4]	0 [0, 1]	1 [1, 7]	NS
Current weight (grams)	720 [539, 917]	720 [515, 970]	680 [565, 865]	764 [513, 1010]	NS
Postmenstrual age (weeks)	25.8 [23.9, 27.9]	26 [24.7, 28.1]	24 [23.4, 26.7]	25.8 [24, 28.5]	NS
Ventilation mode					
CPAP/ Non-invasive NAVA	6 (13.3)	3 (20)	0 (0)	3 (20)	0.03
Conventional MV	1 (2.2)	0 (0)	1 (6.7)	0 (0)	
HFJV	38 (84.4)	12 (80)	14 (93.3)	12 (80)	
Mean airway pressure	7 [6, 9]	7 [6, 10]	8 [7, 10]	6 [6, 7]	0.04
Fraction of inspired oxygen	26 [21, 35]	26 [24, 35]	28 [26, 92]	21 [21, 33]	0.07
Respiratory severity score	1.89 [1.47, 2.96]	1.96 [1.56, 2.2]	2.24 [1.89, 8]	1.47 [1.26, 2.8]	0.02
pH	7.34 [7.31, 7.37]	7.34 [7.31, 7.35]	7.32 [7.11, 7.33]	7.37 [7.34, 7.4]	0.001
Partial pressure of carbon dioxide	44 [41, 53.5]	49 [42, 53]	45 [40, 71]	43 [37, 50]	NS
Oxygenation index	4.13 [2.68, 6.24]	3.98 [2.95, 5.26]	6.2 [4.4, 16.8]	2.84 [2.3, 5.6]	0.008
Base excess/deficit	-1.22 ± 4.74	-2.62 ± 4.08	-4.36 ± 4.78	.5 ± 3.85	0.002
Echocardiographic features					
Presence of LV dysfunction	4 (8.9)	0 (0)	4 (26.7)	0 (0)	0.01
Presence of RV dysfunction	8 (17.8)	0 (0)	7 (46.7)	1 (6.7)	0.001
Atrial communication (%)	44 (97.8)	15 (100)	15 (100)	14 (93.3)	NS
Left-to-right shunt	24 (53.3)	14 (93.3)	4 (26.7)	6 (40)	0.003
Bidirectional/right-to-left shunt	20 (44.4)	1 (6.7)	11 (73.3)	8 (53.3)	
PDA size	1.3 ± 0.97	2.1 ± 0.5	1.7 ± 0.6	–	0.07[a]
PDA size indexed to weight in kilograms	1.95 [0.5, 2.8]	2.6 [2.1, 3.3]	2.3 [1.6, 2.9]	–	NS[b]
Diastolic flow reversal in the post-ductal descending aorta (%)	14 (31.1)	12 (80)	1 (6.7)	0 (0)	<0.001
Modified PDA score[23]	4.6 ± 3.4	9 ± 1	2.8 ± 1.5	–	<0.001[a]
Percentage of time PDA shunt spent right-to-left	–	–	31.5 [17.8, 41.1]	–	–

Results are presented in mean±SD, median [IQR], n (%).
Respiratory severity score was calculated as mean airway pressure × fraction of inspired oxygen.
Abbreviations: CPAP, continuous positive airway pressure; NAVA, neutrally adjusted ventilatory assist; MV, mechanical ventilation; HFJV, high frequency jet ventilation; LV, left ventricle; RV, right ventricle; PDA, patent ductus arteriosus.
[a]Student t-test comparing PDA left-to-right versus PDA bidirectional.
[b]Mann-Whitney test comparing PDA left-to-right versus PDA bidirectional.

were performed using intra-class coefficient. Agreement between investigators was evaluated using the Bland-Altman method to calculate the bias (mean difference) and the 95% limits of agreement 1.96 SD around the mean difference).[25] Bland-Altman graphs were also created to demonstrate agreement statistics between two methods of SVC flow for the entire cohort and for each sub-group. Sub-group analysis was performed for the patients with hemodynamically significant PDA. Results were considered significant if $p < 0.05$. Data were analyzed using SPSS version 26 statistical software [IBM, Armonk, NY, USA]. Although the relationship between LVO and SVC flow has been studied,[3,13,14,17] the influence of PDA shunt directionality or shunt magnitude has not been evaluated. For this study no power calculations were performed and we used a convenience sample size.

3 | RESULTS

Forty-five patients fulfilled inclusion criteria and were included in the study. **Table 1** depicts demographic information and baseline clinical TnECHO characteristics between the three groups. There were four patients identified with small VSDs [three in the left-to-right and one in the bidirectional PDA groups, respectively], all of whom had left-to-right shunts. All infants included in the cohort had forward diastolic

TABLE 2 Comparison of measures of systemic blood flow in patients with different patterns of ductal shunt

	Cohort ($n = 45$)	PDAleft-to-right ($n = 15$)	PDA bidirectional ($n = 15$)	No PDA ($n = 15$)	p
Heart rate (bpm)	159.2 ± 11.7	162.7 ± 12.7	159.8 ± 12.4	155.1 ± 9.1	NS
Left ventricular measurements					
Left ventricular output (ml/min/kg)	186.7 ± 61.3	250.1 ± 45.7	140.4 ± 30.3	169.7 ± 43.7	<0.001
LV VTI (cm)	9.2 ± 2.9	12.3 ± 2.4	7.5 ± 1.4	7.8 ± 1.8	<0.001
LV VTI × heart rate / weight	2072 ± 812	2812 ± 732	1715 ± 643	1691 ± 496	<0.001
Left ventricular outflow tract diameter (cm)	.34 ± .05	.34 ± .04	.33 ± .05	.36 ± .05	NS
Low LVO (< 150 ml/min/kg)	14 (31.1)	0 (0)	8 (53.3)	6 (40)	0.005
High LVO (> 250 ml/min/kg)	7 (15.6)	7 (46.7)	0 (0)	0 (0)	<0.001
Right ventricular measurements					
Right ventricular output (ml/min/kg)	150.6 ± 44.3	161.1 ± 37	120.5 ± 42.2	170.3 ± 39.1	0.003
RV VTI (cm)	6.7 ± 2	7 ± 1.6	5.5 ± 1.8	7.8 ± 1.9	0.004
RV VTI × heart rate / weight	1557 ± 595	1684 ± 646	1286 ± 541	1701 ± 533	0.09
Right ventricular outflow tract diameter (cm)	.36 ± .05	.36 ± .04	.36 ± .06	.36 ± .05	NS
Modified method for SVC flow					
Superior vena cava flow modified (ml/min/kg)	69.4 ± 29.5	65 ± 35.1	69.7 ± 24.7	73.4 ± 29.4	NS
SVC suprasternal VTI (cm)	4.7 ± 1.6	4.1 ± 1.6	4.8 ± 1.8	5.1 ±1.5	NS
SVC mean CSA (mm^2) from suprasternal view	6.8 ± 2.3	6.6 [4.3, 9.2]	5.8 [5.6, 8.5]	6.9 [4.2, 7.5]	NS
SVC VTI × heart rate/ weight (modified method)	1057 ± 431	957 ± 394	1049 ± 374	1166 ± 514	NS
Low modified SVC flow (< 60 ml/min/kg)	19 (42.2)	8 (53.3)	6 (40)	5 (33.3)	NS
Traditional method for SVC flow					
Superior vena cava flow traditional (ml/min/kg)	106.2 ± 48.1	102.2 [74.8, 129.6]	88.9 [56.1, 123.8]	107.6 [81.8, 147.5]	NS
SVC subcostal VTI (cm)	8.4 ±2.7	7.8 ± 2.5	8.4 ± 2.8	9 ± 2.8	NS
SVC mean diameter (mm)	2.7 ±0.5	2.8 [2.2, 3]	2.3 [2.2, 3]	2.8 [2.2, 3.1]	NS
SVC CSA (mm^2) calculated from diameter	5.5 [4. 7]	4.9 [3.9, 7.2]	6.5 [4.2, 7.2]	5.1 [3.7, 6.6]	NS
SVC VTI × heart rate / weight (traditional method)	1729 [1368, 2225]	1816 [1164, 2066]	1689 [1371, 2290]	1729 [1411, 2235]	NS
Low traditional SVC flow (< 60 ml/min/kg)	9 (20)	2 (13.3)	4 (26.7)	3 (20)	NS
Ratios					
LVO/RVO	1.16 [0.96, 1.53]	1.44 [1.34, 1.77]	1.19 [0.9, 1.55]	0.98 [0.93, 1.1]	<0.001
LVO/Modified SVC flow	2.54 [2.01, 3.65]	3.59 [3.03, 6.62]	2.05 [1.39, 2.54]	2.35 [2.04, 3.26]	0.001
LVO/Traditional SVC flow	1.74 [1.35, 2.4]	2.26 [1.7, 4.19]	1.38 [1.15, 2.23]	1.65 [1.13, 2.28]	0.013

Results are presented in mean±SD, median [IQR], n (%).
Abbreviations: PDA, patent ductus arteriosus; LV, left ventricle; VTI, velocity time integral; LVO, left ventricular output; RV, right ventricle; RVO, right ventricular output; SVC, superior vena cava; CSA, cross-sectional area.

flow to the MCA. **Table 2** depicts the different methods of estimation of systemic blood flow between the three groups. SVC dimensions and flow measurements were not different between the groups regardless of the technique utilized, even though patients with left-to-right shunts had higher LVO/RVO compared to the other groups [**Table 2**].

Correlation between different methods of SVC flow: **Figure 1** depicts the agreement statistics between both methods of SVC flow for the entire cohort and for each of the subgroups. There was strong positive correlation between both methods of SVC flow [Pearson's correlation .5, $p < 0.001$], SVC VTI [Pearson's correlation .61, $p < 0.001$], SVC CSA [Pearson's .51, $p < 0.001$] and SVC VTI × HR / weight [Pearson's correlation .7, $p < 0.001$].

Correlation between SVC flow and other indices of systemic blood flow: The correlation between SVC flow and LVO was weak in patients with significant left-to-right PDA shunt, irrespective of the method utilized [**Table 3**]. The correlation between SVC flow and LVO was strong using the modified method, but weak with the traditional method, in patients with no transductal shunt. For the overall cohort and for patients with bidirectional shunts, the correlations between LVO and SVC flow were overall poor [**Table 3**, **Figure 2**]. There was a strong correlation between traditional SVC flow and RVO in patients with left-to-right shunt [**Table 3**]. The correlation between SVC VTI × HR/ weight and LV VTI × HR/weight was small to moderate for the entire cohort, but strong for the patients with no PDA [**Table S2**]. The correlation

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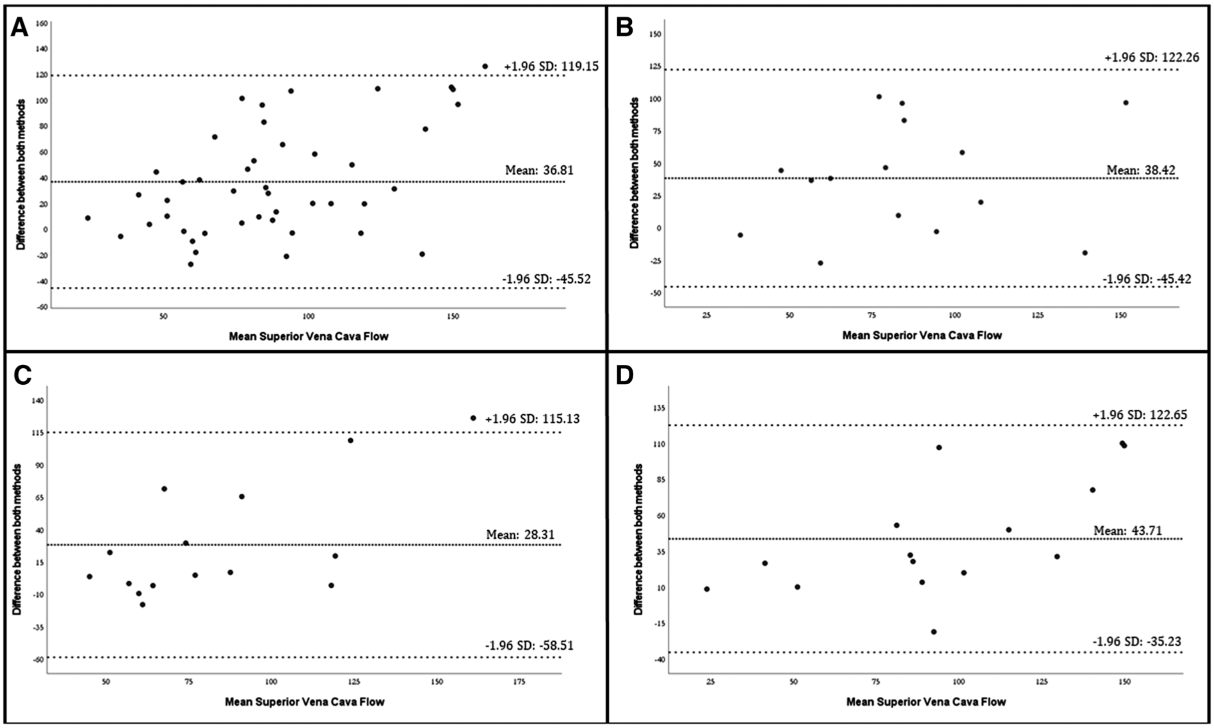

FIGURE 1 Bland-Altman plot for both methods of superior vena cava flow showing mean percentage line (fine dotted line) and 95% limits of agreement (thick dotted lines) for the entire cohort (A), patients with left-to-right shunt through patent ductus arteriosus (PDA) (B), patients with bidirectional PDA (C) and patients with no PDA (D)

TABLE 3 Pearson's correlation coefficient between superior vena cava flow and cardiac outputs in different patient populations

	Left ventricular output	*p*-value	Right ventricular output	*p*-value	LVO/RVO ratio	*p*-value
SVC flow (modified method)	.144	NS	.359	0.015	-.199	NS
SVC flow (traditional method)	.183	NS	.391	0.008	-.242	NS
Superior vena cava flow (modified method)						
PDA left-to-right	.109	NS	.323	NS	-.250	NS
PDA Bidirectional	.301	NS	.292	NS	-.176	NS
No PDA	.630	0.012	.635	0.011	.159	NS
Superior vena cava flow (traditional method)						
PDA left-to-right	.155	NS	.490	0.064	-.450	0.092
PDA Bidirectional	.476	0.073	.397	NS	-.183	NS
No PDA	.283	NS	.269	NS	.087	NS
Left ventricular output						
PDA left-to-right	–	–	.516	0.049	.198	NS
PDA Bidirectional	–	–	.511	0.052	.054	NS
No PDA	–	–	.936	<0.001	.447	0.095

Abbreviations: SVC, superior vena cava; PDA, patent ductus arteriosus; VTI, velocity integral time; HR, heart rate.

between SVC VTI × HR/ weight and RV VTI × HR/weight was moderate for the entire cohort, but strong for the patients with no PDA [Table S2].

Correlation between LVO and RVO: The correlation between LVO and RVO was strongly positive for the entire cohort [Pearson's .581, $p < 0.001$], with highest correlation in patients with no PDA [**Table 3**]. There was also a strong correlation between LV VTI × HR / weight and RV VTI × HR / weight [Pearson's correlation .707, $p < 0.001$] which was highest in patients with left-to-right shunts, followed by no PDA [Table S3].

Low LVO and low SVC flow: Low LVO was more common in patients with bidirectional shunts or no PDA, whereas low SVC flow was equally distributed amongst the three groups [**Table 2**]. Low modified SVC flow was present in nine (64.3%) patients with low LVO ($p = 0.09$) while low

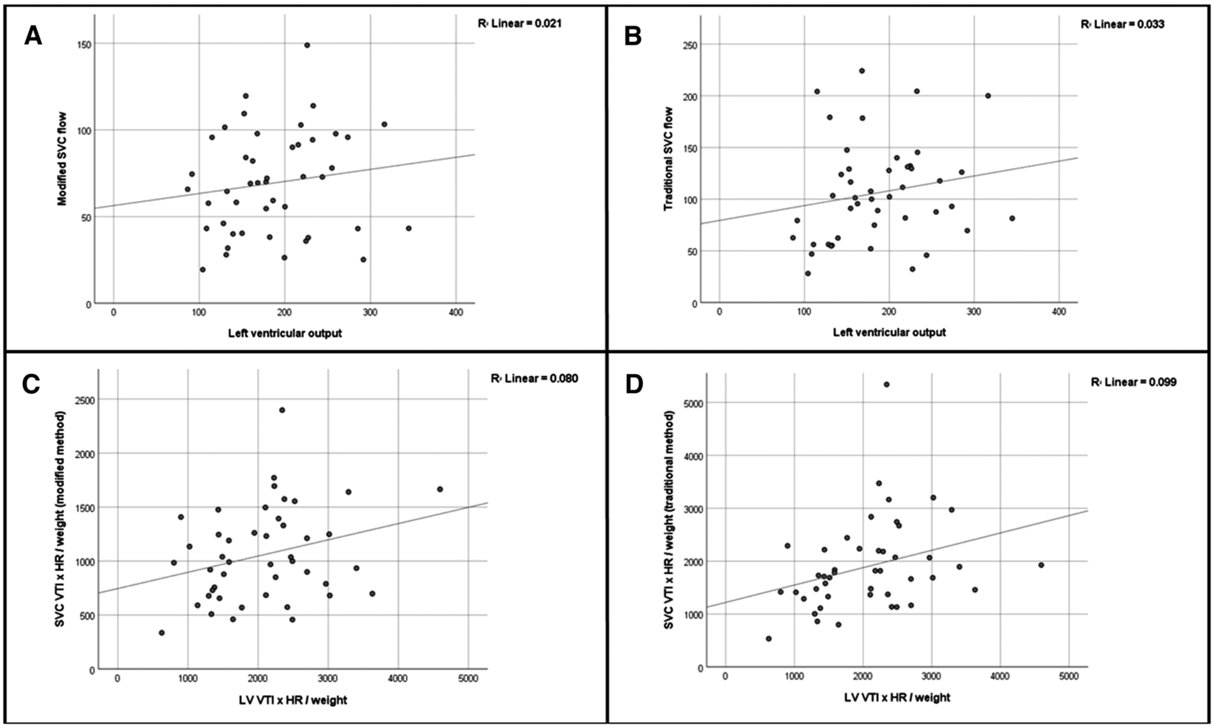

FIGURE 2 Relationship between left ventricular output and superior vena cava flow for the entire cohort. (A) Modified superior vena cava flow versus LVO; (B) Traditional superior vena cava flow versus LVO; (C) Modified SVC × HR / weight versus LV VTI × HR / weight; (D) Traditional SVC × HR / weight versus LV VTI × HR /weight

traditional SVC flow was present in six (42.9%) patients with low LVO ($p = 0.03$).

Intra- and Inter-observer reliability: Intra-observer reliability was strong for all measurements but with broader limits of agreement for the traditional SVC flow [Figure S1]. Intra-observer reliability intra-class coefficient values were: LVO .99, RVO .96, traditional SVC flow .95 and modified SVC flow .97. Inter-observer reliability coefficients were: LVO .96, RVO .9, traditional SVC flow .81 and modified SVC flow .86. Inter-observer reliability was worse for SVC diameter/ calculated CSA, and consequently for traditional SVC flow measurements [Table S3, Figure S2].

SVC flow and hemodynamic significance of PDA: Sub-analysis of the patients with hemodynamically significant left-to-right PDA shunts showed no correlation between PDA score[23] and SVC flow with either method [Figure S3]. There was no correlation between PDA score and LVO/SVC flow ratio (Pearson's correlation .30, $p = 0.27$). Similarly, there was no association between PDA size indexed to weight and SVC flow [data not shown]. We found no correlation between high LVO and low SVC flow in this group (modified SVC method $p = .62$, traditional SVC method $p = 0.47$).

4 | DISCUSSION

In this cohort of preterm infants, we found that the correlation between SVC flow and other estimates of systemic blood flow (LVO and RVO) was weak in the presence of a hemodynamically significant PDA. These findings refute our hypothesis that pre-ductal cardiac output (LVO) would increase SVC flow. Additionally, irrespective of significant differences in LVO and RVO according to shunt directionality, SVC flow was relatively maintained across all three groups. Miletin et al. similarly found that in a cohort of preterm infants where the majority of patients had transductal shunts, SVC flow did not correlate with LVO nor RVO.[26] Furthermore, in previous reports SVC flow correlated with LVO only in patients in no PDA.[3] In our cohort, however, the strong correlation between LVO and SVC flow was significant only when using the modified SVC flow method but not the traditional method. Our findings raise concerns with the reliability of estimating SVC flow specifically by using the traditional method. Contrary to previous reports,[3] the traditional method had a weak correlation with other estimates of systemic blood flow in patients with hemodynamically significant left-to-right or no PDA shunt. SVC flow appears to be preserved in this cohort of patients, irrespective of shunt directionality and of changes in cardiac output due to the presence of a PDA. These findings can be explained by either inherent methodological concerns or compensation for left-to-right ductal shunt with increased LVO to maintain systemic perfusion. The impact of prolonged PDA shunt on LVO, SVC flow, and cerebral flow remains unknown.

Echocardiography assessment of both LVO and SVC flow have been previously validated with cMR; specific findings revealed that LVO is more accurate than SVC flow.[13] Our study corroborates the fact that LVO has better intra- and inter-observer reliability than SVC flow. Furthermore, we found that irrespective of the presence or directionality of PDA shunt, there is inconsistent bias based on the mean average for the cohort. The reliability of SVC flow is mainly affected by inaccuracies in obtaining diameter/cross-sectional area

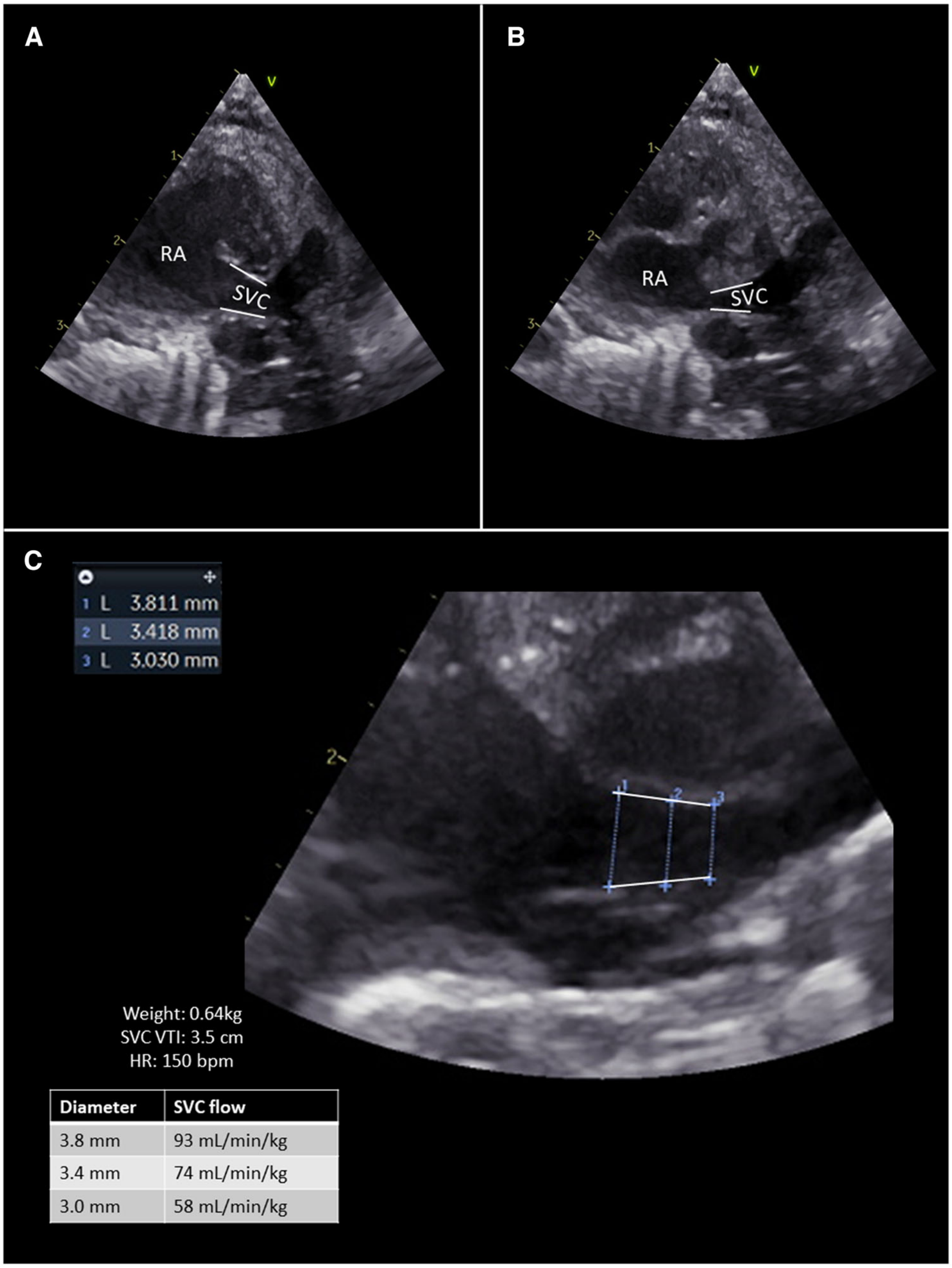

FIGURE 3 Superior vena cava (SVC) view from a parasternal long axis view with the beam in a true sagittal plane and angled to the right of the ascending aorta. The opening of the SVC into the right atrium frequently assumed a funnel-shape that changes according to the phase of the cardiac cycle (A and B). Small variation in vessel diameter estimation can have a significant impact on SVC flow calculations (example C)

of the vessel.[11,14,27] Detailed methodology and landmarks have been previously described.[3,14] In the traditional method, the measurement of SVC diameter lacks discrete landmarks such as the hinge points of the aortic and pulmonary valves, which increases the likelihood of inconsistency. Asymmetry in vessel diameter, variability in the caliper of the vessel depending on the phase of the cardiac cycle and a crescent-shaped cross-section increases the predisposition to measurement error.[13] The SVC opening into the right atrium frequently assumes a funnel shape, such that, it is difficult to determine the most accurate place for measurement. Squaring the radius to obtain cross-sectional area increases the variability. **Figure 3** illustrates the changes in SVC shape depending on the phase of the cardiac cycle and provides an example as to how even small variations in vessel diameter can result in significant changes in the estimation of SVC flow, particularly in patients with a low weight [small denominator]. The use of M-mode tracing may improve intra- and inter-observer reliability[3]; however, test-retest accuracy is likely to be impacted in a similar fashion if the tracing is placed slightly more inward or outwards where the vessel has a funnel-shaped opening towards the right atrium [**Figure 3**]. Furthermore, even with higher repeatability amongst other groups, significant variation in the normative data derived from different centers exists.[3,17,28–32] The modified SVC flow method had better intra- and inter-rater reliability, similar to what has been validated with cMR imaging. This is primarily due to improved estimation of the cross-sectional area of the vessel.[14] The modified method also has the advantage of less interference by respiratory movements changing the plane of insonation and lack of abdominal air degradation of image quality. The VTI appears to be underestimated in both methods, which is most likely due to non-laminar flow in the vessel.[14] Comparison between flow methods irrespective of vessel/annulus size showed improved correlation, albeit modest for patients with transductal shunts.

Patients with left-to-right PDA shunt had higher LVO/RVO ratio, suggestive of systemic steal phenomenon, while patients with bidirectional shunts or no PDA had LVO/RVO closer to one. Although LVO/SVC ratio was higher in patients with left-to-right PDA shunts, this was not related to significant differences in SVC flow, but rather due to higher LVO in comparison to the other groups. Previous studies correlated low SVC flow and higher LVO/SVC ratio with increased diameter of PDA in preterm infants.[5,33] None of these studies, however, reported changes in SVC flow alone in relation to ductal diameter, and no other markers of shunt volume were assessed. On subgroup analysis we also did not find any association between SVC flow and PDA score, nor with PDA diameter indexed to weight. While it is plausible that auto-regulatory mechanisms are playing a modulator role in maintaining cerebral flow perfusion despite the presence of a hemodynamically significant left-to-right PDA, we cannot ascertain whether the effects of chronic shunts and/or longer exposure on cerebral vessel remodeling would have a differential impact on cerebral blood flow. Importantly, our cohort included young infants (most less than 7 days old) and none of the patients had evidence of abnormal (absent/reversed) MCA diastolic flow. This is similar to other studies that show abnormal MCA flow, in the setting of PDA, occurs predominantly with prolonged shunts.[34–36] Our study did not evaluate the relationship between cerebral perfusion using MCA blood flow velocity, resistive index, or other markers such as near infrared spectroscopy.[37] Even though LVO is not truly reflective of systemic blood flow, in the presence of left-to-right PDA shunt, the concomitant evaluation of LVO and RVO and careful assessment of transductal and intra-cardiac shunts may provide insights related to systemic blood flow.

This study has several limitations. *First*, we had a small sample size with a relatively wide range of gestational age, weight, and postnatal age. It has been previously proposed that the repeatability of measures may be different in the transitional period versus later during the NICU course, although this was not validated in a cMR study.[14] *Second*, the tracing of the RV VTI envelope may be challenging in the presence of diastolic flow disturbance caused by left-to-right shunts. Although we assumed a triangular shape of the main pulmonary artery Doppler envelope, inter-observer reliability was worse for RVO when compared to LVO. *Third*, the magnitude of the role of intra-cardiac shunts (i.e., patent foramen ovale) cannot be reliably assessed with echocardiography but could play a role in our findings. In patients with exclusive intra-cardiac shunts, and no PDA, LVO/RVO may provide insights on degree of shunt volume. On the contrary, in patients with left-to-right PDA shunt, the presence of an atrial communication may alter the LVO/RVO ratio whilst being a marker of hemodynamic significance.[23] *Last*, we do not provide any correlation between any of the measurements of systemic blood flow and other markers of circulatory failure [i.e., arterial blood pressure, lactic acid, urinary output] or other neonatal outcomes. Although low SVC flow has been previously associated with intra-ventricular hemorrhage in preterm infants,[5] these findings were reported only in the transitional period. Our study included TnECHOs done beyond the transitional period as well and, therefore, a similar association cannot be ascertained.

5 | CONCLUSION

SVC flow was maintained irrespective of increased pre-ductal cardiac output in patients with left-to-right shunt. This may reflect autoregulation of cerebral perfusion and/or concerns with the methodology of SVC flow estimation. In addition, both intra- and inter-observer reliability are worse for SVC flow when compared to LVO and RVO. While it is physiologically clear that LVO is not a reliable measure of post-ductal systemic blood flow, the contrary should also not be interpreted; specifically, that SVC flow is a superior measure of systemic blood flow in the presence of a PDA. Further prospective studies should be performed to assess the correlation between markers of systemic blood flow, relevant neonatal outcomes and other indices of adequacy of systemic perfusion according to different patterns and variable durations of ductal shunt. More advanced imaging methods, such as cMR imaging or cardiac catheterization, may help delineate patterns among immature infants according to different patterns of intra- and extra-cardiac shunts.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to disclose.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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